$$W_2 = \frac{15}{16}E^2C \qquad (16)$$

1.

$t = 0$ D_1 $u = U_0\sin\left(\frac{2\pi}{T}t\right)$ C_1

$u = 0$ C_1 C_1

C_2 $t = T/2$ $T/2$ C_2

C_2 C_1 C_2

v_n

$$\frac{l}{v_n} = n\frac{T}{2} \quad n = 1,2,\cdots \qquad (1)$$

C_1 $t = 0$ C_2

$$v_n = \frac{2l}{nT} \quad n = 1,2,\cdots \qquad (2)$$

C_1 C_2 v_n

$$\Delta t_{n1} = \frac{1}{2}nT \qquad (3)$$

C_2 G

$$\Delta t_{n2} = \frac{l}{2v_n} \qquad (4)$$

G

$$t_n = \Delta t_{n1} + \Delta t_{n2} = \frac{3}{4}nT \qquad (5)$$

D_2 $t = 10T/4$

$$t_n \leqslant \frac{10}{4}T \qquad (6)$$

C (5)、(6) n

n

$$n_{max} = 3 \qquad (7)$$

$$E_n = \frac{1}{2}mv_n^2 + E_0\sin\left[\frac{2\pi}{T}\left(\frac{3n}{4}T\right) + \frac{\pi}{4}\right]$$

$$n = 1,2,3 \qquad (8)$$